

Unnamed_Unit

Unit 3 - Wishing Candle Creation

PDF Documents

PDF Document 1: 1738219072734-Chapter 9 - Unit 3 - Wishing Candle Creation.pdf

This PDF document is included in the unit materials.



UNIT 3 WISHING CANDLE CREATION

BLOWING OUT CANDLES



A common tradition is to put candles on a birthday cake. The birthday person makes a wish and blows out the candles. The Significance of Wishes

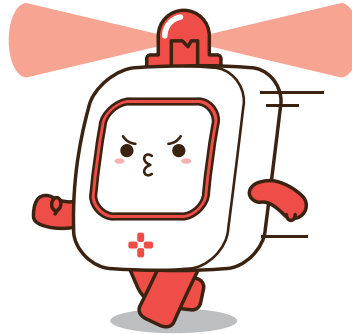
WHY DO WE MAKE WISHES?

It's a fun and magical part of the birthday celebration. People believe your wish will come true if you blow out all the candles in one breath.



DO YOU REMEMBER WHAT ENVIRONMENT DO?

It can sense four types of things!



Environment

TEMPERATURE Measures how hot or cold the surroundings are.

HUMIDITY Checks the amount of moisture in the air.

LIGHT Gauges the brightness of the light around it.

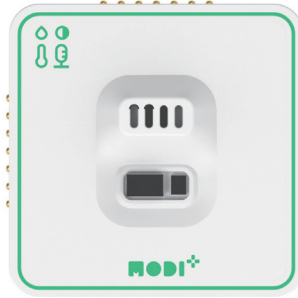
VOLUME Detects the level of sound in the environment.

Activity: Wishing Candle

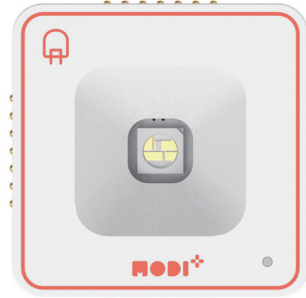
1. Materials Needed:



Network



Environment

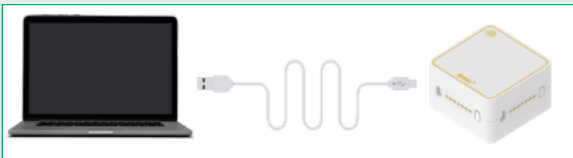


LED

2. Connect the Network with computer

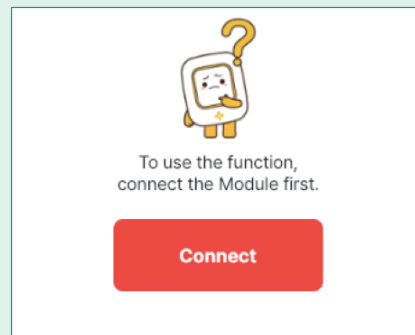
Step 1

Connect the Network Module to your device using a Type-C cable.



Step 2

Click the connect button in the module map.



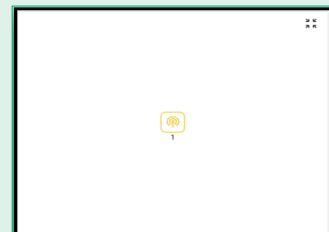
Step 3

Select 'MODI+ Network Module-Paired' and then click 'Connect' again.



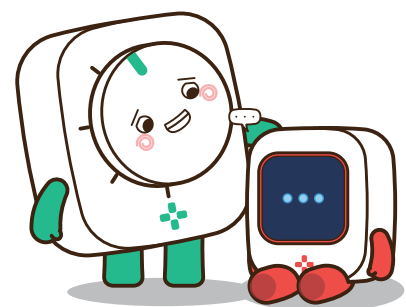
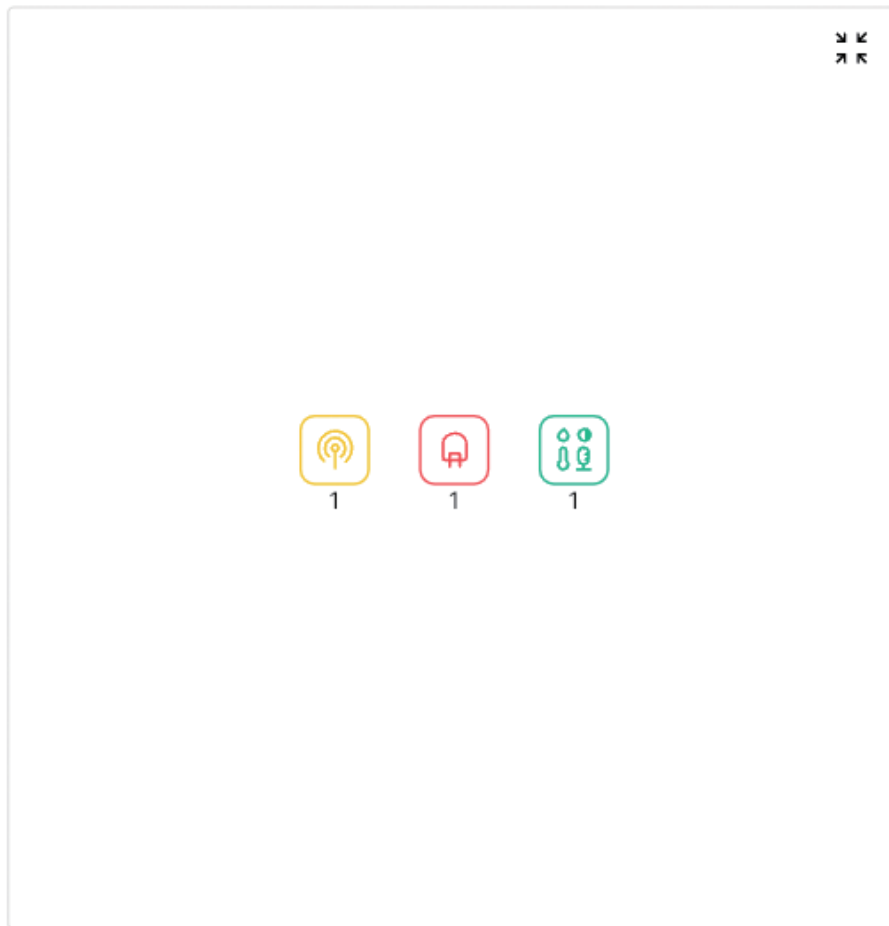
Step 4

A network icon will appear in the Module Map when the connection is successful.



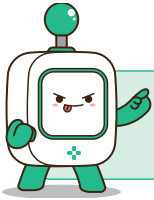


- 3. Let's attach the Environment and LED modules next to the Network! Once everything is connected, you should see three modules displayed on the module map.**



Coding

Code Story



Topic Camp Fire



DEFINITION

CODE DESIGN



ABSTRACTION

CODE DESIGN

We are on camp!
Let's Light it up

How should it work?
Shall we write down the functions in order?



Fire is off



Blow the wood



Fire is on for 2 seconds



Fire is off



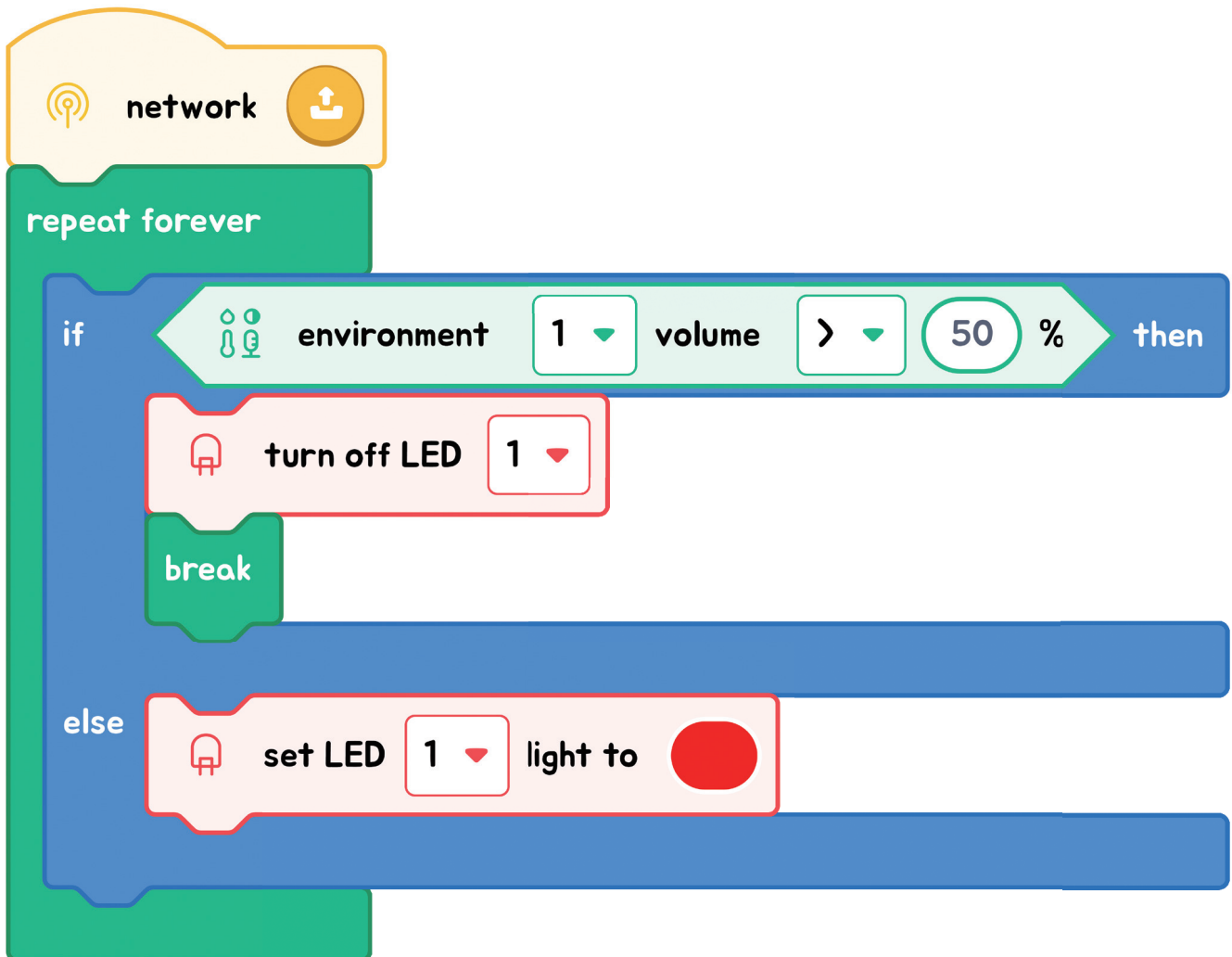
DECOMPOSITION

STORY TO CODE

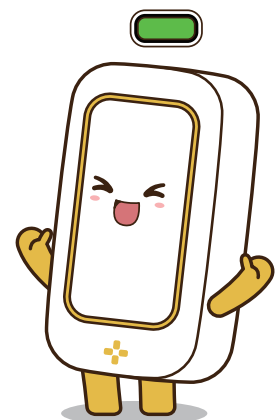
To make the campfire flames flicker or rise by blowing, what module would be needed?
Please write down the names of the modules you can use.



Code Quest

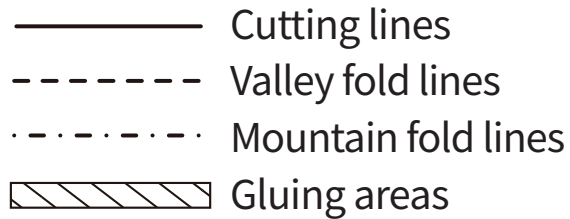
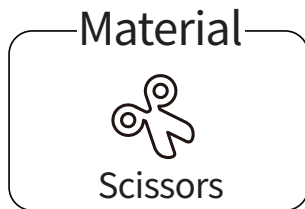


Now, let's use the template to create a birthday cake! Combine the template and the modules well to make the candles together.



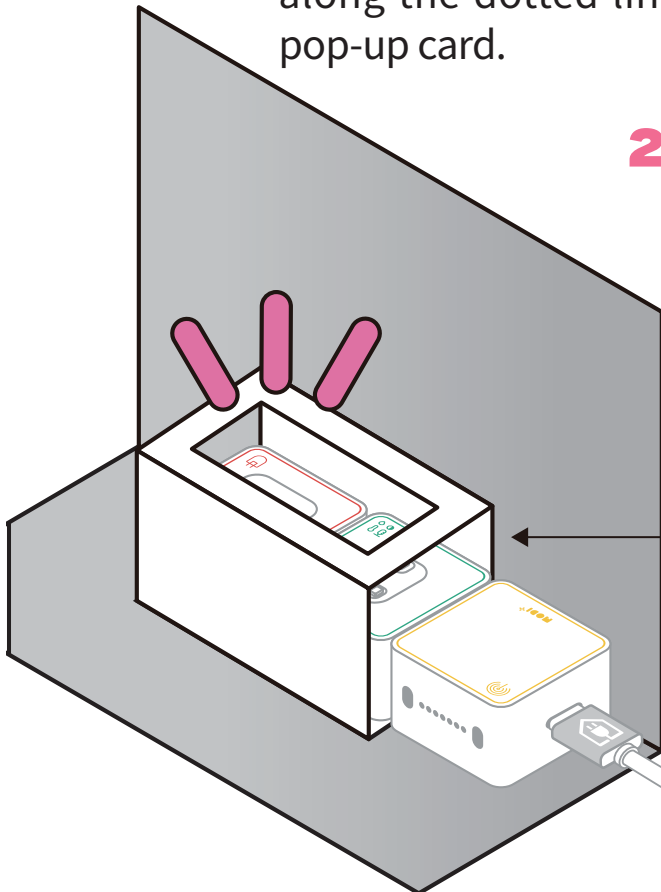
Birthday Cake

Cut along the lines and fold to make the cake.



Designer MODI

- 1** Cut along the solid lines and fold along the dotted lines to create a pop-up card.



- 2** As shown in the illustration, Assemble the LED, environment, and battery module and place them in the space for the cake's candles.

